

INNOTRACK INTERNSHIP PROGRAM - 2026



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- **Project Name – AIR QUALITY INDEX PREDICTOR**
- **Theme – MACHINE LEARNING IN SMART CITY**
- **VTU ID – 28149**



AIR QUALITY INDEX PREDICTOR



INTRODUCTION :

Air pollution is one of the most significant environmental challenges faced by modern cities. Poor air quality directly affects human health, causing respiratory diseases, cardiovascular problems, and reduced quality of life. Monitoring air quality manually can be difficult due to the large amount of environmental data generated every day.

The Air Quality Index Predictor is a Machine Learning-based system designed to predict air quality levels using environmental parameters such as PM2.5, PM10, Carbon Monoxide (CO), Nitrogen Dioxide (NO₂), Sulfur Dioxide (SO₂), Ozone (O₃), temperature, and humidity. The system helps government authorities, environmental agencies, and citizens understand air pollution trends and take preventive measures. This project contributes to smart city development by enabling data-driven environmental monitoring and decision-making.



Idea and solution:

IDEA

The idea behind this project is to develop an intelligent system that can predict Air Quality Index levels using historical and real-time environmental data. By leveraging machine learning algorithms, the system can identify pollution patterns and forecast air quality conditions efficiently.

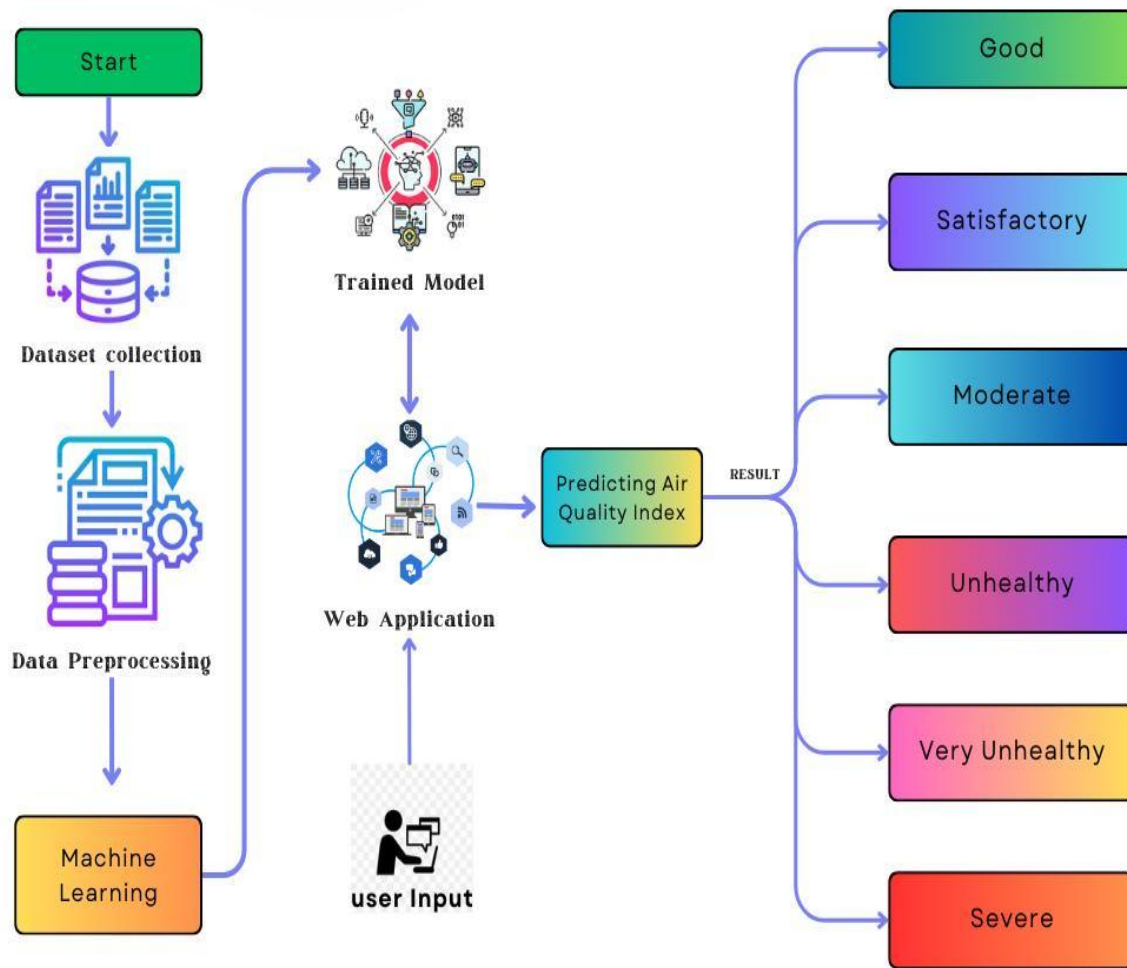
SOLUTION

The proposed solution is a web-based Air Quality Index Predictor developed using Python, Scikit-learn, Flask, and SQLite. Users can enter environmental parameters through a simple interface, and the trained machine learning model predicts the AQI category instantly. The application provides a user-friendly platform for monitoring environmental conditions and generating meaningful air quality insights.

Uniqueness:

- *Predicts Air Quality Index using Machine Learning techniques.*
- *Supports real-time environmental monitoring.*
- *Provides fast and automated AQI prediction.*
- *Helps citizens understand pollution levels.*
- *Assists government agencies in environmental planning.*
- *Combines data analytics, machine learning, and web technology.*
- *User-friendly and easy-to-access web interface.*
- *Supports smart city sustainability initiatives.*
- *Generates meaningful insights from environmental data.*
- *Can be integrated with future IoT-based smart monitoring systems.*

TECHNICAL APPROACH



Technologies to be Used:

Python was used as the primary programming language for implementing machine learning algorithms and handling environmental datasets. Pandas and NumPy were utilized for data preprocessing, cleaning, and numerical analysis. Scikit-learn was used to train, test, and evaluate machine learning models for AQI prediction. Flask was used to deploy the trained model as a web application. SQLite was used for storing environmental records and prediction results. HTML, CSS, and JavaScript were used to create an interactive and responsive user interface. Matplotlib was used for visualization and exploratory data analysis.

Methodology:

Dataset Collection, Data Cleaning and Preprocessing, Exploratory Data Analysis, Feature Selection, Model Training, Model Evaluation, AQI Prediction, Model Saving (.pkl), Database Integration, Flask Web Deployment

FEASIBILITY AND VIABILITY



Technical Feasibility :

The project utilizes established machine learning algorithms and open-source technologies, making implementation practical and efficient.

Economic Feasibility :

Development costs are minimal as all tools and software libraries are freely available.

Operational Feasibility :

The application is easy to use and can be accessed through a simple web interface by citizens and environmental authorities.

Viability :

- *Supports smart city development.*
- *Improves environmental awareness.*
- *Assists government decision-making.*
- *Easy to maintain and update.*
- *Scalable for large cities.*
- *Promotes healthier living environments.*

CHALLENGES FACED



During the development of the Air Quality Index Predictor, several challenges were encountered. One of the major challenges was obtaining a clean and consistent environmental dataset. The dataset contained missing values, duplicate records, and inconsistencies that required extensive preprocessing.

Another challenge involved selecting the most suitable machine learning algorithm. Multiple algorithms were tested and compared to determine the model that delivered the highest prediction accuracy. Hyperparameter tuning was also performed to improve model performance and reliability.

Integrating the machine learning model with the Flask web application required careful implementation to ensure seamless communication between the frontend, backend, and database. Additional challenges included dependency management, deployment issues, and validating prediction accuracy under different environmental conditions.

FUTURE ENHANCEMENTS



The Air Quality Index Predictor can be enhanced by integrating real-time IoT sensors for live environmental data collection. Advanced machine learning algorithms such as XGBoost, Random Forest, Gradient Boosting, and Deep Learning models can be incorporated to improve prediction accuracy.

Future versions of the system may include interactive dashboards that visualize pollution trends using charts, maps, and graphs. Mobile application support can improve accessibility and user engagement. Cloud deployment can further enhance scalability and availability.

Additional enhancements may include:

- *GPS-based location-specific AQI prediction.*
- *Pollution alert notification system.*
- *Smart city analytics dashboard.*
- *Weather-based pollution forecasting.*
- *Integration with government environmental databases.*
- *AI-based recommendations for pollution reduction.*
- *Automated reporting and analytics generation.*
- *Public health risk assessment module.*

IMPACT AND BENEFITS

IMPACTS :

- *Improves public health by providing early warnings about harmful air pollution levels and environmental risks.*
- *Supports smart city development through intelligent monitoring and analysis of air quality conditions.*
- *Enables government authorities to take proactive actions for pollution control and urban environmental management.*
- *Promotes data-driven decision-making by converting environmental data into meaningful AQI predictions.*
- *Increases public awareness about air quality and encourages citizens to adopt healthier lifestyles.*
- *Contributes to environmental sustainability by helping reduce the long-term impact of air pollution.*

BENEFITS :

- *Provides fast and accurate Air Quality Index predictions using advanced Machine Learning algorithms.*
- *Helps citizens plan outdoor activities safely by understanding current and future air quality conditions.*
- *Assists environmental agencies in monitoring pollution trends and implementing effective control measures.*
- *Reduces manual effort involved in analyzing large volumes of environmental and pollution data.*
- *Offers a simple and user-friendly web interface that can be easily accessed by both citizens and authorities.*
- *Supports real-time environmental monitoring and enhances smart city infrastructure capabilities.*
- *Creates a scalable platform that can be expanded with IoT sensors, cloud services, and mobile applications.*
- *Improves overall quality of life by enabling cleaner, healthier, and safer urban environments.*

CONCLUSION

- *Successfully developed a Machine Learning-based Air Quality Index Predictor.*
- *Applied data preprocessing and feature engineering techniques.*
- *Trained and evaluated machine learning models for AQI prediction.*
- *Achieved reliable and accurate air quality prediction results.*
- *Integrated the prediction model into a Flask web application.*
- *Utilized SQLite database for efficient data management.*
- *Demonstrated the practical use of Machine Learning in Smart City solutions.*
- *Provided a scalable platform for environmental monitoring.*
- *Improved awareness regarding air pollution and public health.*
- *Established a strong foundation for future smart environmental systems.*

THANK YOU